



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE

(AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation:R23		IV/ IV-B.Tech. I-Semester							
CIVILENGINEERING									
COURSESTRUCTURE									
(Witheffectfrom2023-24admittedBatchonwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CE4101	Geotechnical Engineering -II	PC	3	0	0	3	30	70	100
B23HS4103	Advanced Construction Management	HS	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23CE4112	3D Building Modelling	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	0	30	-	30
B23CE4113	Evaluation of Industry Internship /Mini Project.	PR	--	--	--	2	--	50	50
TOTAL			19	1	2	21	240	540	780

	Course Code	Course
#PE-IV	B23CE4102	Prestressed Concrete
	B23CE4103	Advanced Environmental Engineering
	B23CE4104	Urban Hydrology
	B23CE4105	Foundations on Expansive Soils
	B23CE4106	MOOCS-IV (12-week Swayam/NPTEL course recommended by the BoS)
#PE-V	B23CE4107	Geosynthetics and its applications
	B23CE4108	Environmental Impact assessment
	B23CE4109	Disaster Management and Preparedness
	B23CE4110	Traffic Engineering Management
	B23CE4111	MOOCS-V (12-week Swayam/NPTEL course recommended by the BoS)
#OE-III	Student has to study one Open Elective offered by AIDS or AIML or CIC or CSBS or CSG or CSE or CSIT or ECE or EEE or ME or IT or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by AIDS or AIML or CIC or CSBS or CSG or CSE or CSIT or ECE or EEE or ME or IT or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4101	PC	3	--	--	3	30	70	3Hrs.
GEOTECHNICAL ENGINEERING II								
(For CE)								
Course Objectives: The objectives of this course are to make the student aware of								
1.	Different techniques of soil exploration.							
2.	Various methods for estimating bearing capacity of different types of foundations.							
3.	Estimating Load capacity of single piles and groups of piles.							
4.	Settlement analysis of foundations.							
5.	Theoretical aspects of well foundations.							
6.	Concepts of stability of finite and infinite slopes.							
7.	Earth pressures on retaining walls by Rankine’s and Coulomb’s theories							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Select appropriate subsoil exploration methods, sampling techniques, and bore log procedures for planning and interpretation of site investigation programs as per standard practice.							K3
2.	Determine bearing capacity and settlement for the design of shallow foundations using Terzaghi and IS methods ensuring safety and serviceability.							K3
3.	Compute load carrying capacity of single piles and pile groups using static and dynamic methods for selection of suitable deep foundation systems under practical conditions.							K3
4.	Evaluate factor of safety using limit equilibrium methods (Swedish circle, friction circle, Fellenius, Taylor’s charts) for different slope conditions.							K3
5.	Estimate earth pressures using Rankine’s and Coulomb’s theories for retaining wall design under various conditions.							K3
SYLLABUS								
UNIT-I (10Hrs)	Sub Soil Exploration: Methods of sub soil exploration Direct, semi direct and in direct methods, Soundings by Standard, Dynamic cone and static cone penetration tests, Types of Boring, Types of samples, Criteria for undisturbed samples, Transport and preservation of samples, bore logs, planning of exploration program, report writing.							
UNIT-II (10 Hrs)	Shallow Foundations: Types of shallow foundations, Factors effecting location of foundation and design considerations of shallow foundations, proportioning of shallow foundations – isolated, rectangular and trapezoidal combined footings. Bearing Capacity: Safe bearing capacity and allowable bearing pressure, General and local shear failures, Terzaghi’s bearing capacity equations its modifications for square, rectangular and circular foundations, Factors affecting bearing capacity of Soil, Effect of							

	water table on bearing capacity, IS Code method for Bearing capacity of footings, Allowable bearing pressure based on N-values, Field plate load tests, Settlement Analysis of shallow foundations.
UNIT-III (10 Hrs)	Pile Foundations: Types, Construction, load carrying capacity of single piles in sands and clays(α-method) Dynamic Formula, Static formula, Pile load tests, Load carrying capacity of pile groups, Settlement Analysis of pile foundations. Caissons: Types of caissons, pneumatic caissons, Different shapes of well foundations, Relative advantages and disadvantages, Different Components of well and their function, Grip length, problems in well sinking and remedial measures.
UNIT-IV (10 Hrs)	Stability Analysis of Slopes: Infinite and Finite Slopes, Stability Analysis of Infinite Slopes, different factors of safety, Types of Slope Failures – Toe, slope and Base failure, Stability Analysis of Finite slopes – Swedish Circle method, Friction Circle method, Fellenius method for location of Critical Slip Circle, Taylor's stability number.
UNIT-V (10 Hrs)	Earth Pressure: Types of Earth pressure, Rankine's Active and passive earth pressure, Smooth Vertical wall with horizontal and inclined backfills. Coloumb's wedge theory, Culmann's active earth pressure.
Textbooks:	
1.	Soil Mechanics and Foundation Engineering– K. R. Arora
2.	Foundation Analysis and Design –J. E. Bowles.
Reference Books:	
1.	Principles of Foundation Engineering- Braja M. Das
2.	Basic and Applied Soil Mechanics by Gopal Ranjan and A. S. R. Rao
3.	Foundation Engineering-P. C. Verghese
e-Resources	
1.	Nptel - https://nptel.ac.in/courses/105105176

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4103	HS	2	--	--	2	30	70	3 Hrs.
ADVANCED CONSTRUCTION MANAGEMENT								
(For CE)								
Course Objectives:								
1.	Understand and apply various project planning and networking techniques such as PERT and CPM for construction projects.							
2.	Analyze time-related parameters in project networks including floats, slack, and project duration.							
3.	Evaluate cost optimization techniques and perform cost–time trade-off analysis in construction projects.							
4.	Apply construction management practices including quality, safety, material, and machinery management.							
5.	Understand and utilize modern tools like BIM and 3D modelling for efficient project management.							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Explain and develop project networks using bar charts, milestone charts, PERT, CPM, and apply Fulkerson’s rule with dummy activities.							K2
2.	Calculate earliest and latest event times, floats, slack, and determine project duration and critical path.							K3
3.	Analyze project costs and evaluate cost–time optimization using crashing techniques, budgeting, and cost–benefit analysis.							K3
4.	Apply construction management practices including quality control, safety procedures, and effective material and machinery management.							K3
5.	Explain and apply BIM and 3D modelling tools to improve project planning, coordination, and efficiency.							K3
SYLLABUS								
UNIT-I (10Hrs)	Project Planning and Networking Techniques: Networking Methods: Bar charts, Milestone charts, PERT and CPM networks – Comparison, Event, Activity, rules for drawing networks, Numbering of events (Fulkerson’s Rule), Dummy activities							
UNIT-II (10 Hrs)	Time Analysis in Networks: Time estimates - Earliest occurrence time -Latest occurrence time -Slack and floats -Project duration -Start and finish time calculations							
UNIT-III (10 Hrs)	Cost Management and Optimization: Cost–Time Optimization -Direct and indirect costs -Total project cost -Cost slopes and crashing techniques -Project updating - Budgeting and cost–benefit analysis							

UNIT-IV (10 Hrs)	Construction Management Practices: Quality management, On-site safety management procedures, Material management, Machinery management, Risk Management.
UNIT-V (10 Hrs)	Modern Construction Management Tools: Building Information Modelling (BIM), 3D modelling applications in construction, Role of BIM in project management, Collaboration and efficiency improvement
Textbooks:	
1.	Construction Project Management: Planning, Scheduling and Controlling by K. K. Chitkara
2.	BIM and Construction Management: Proven Tools, Methods, and Workflows by Brad Hardin and Dave McCool
Reference Books:	
1.	Construction Safety Management, 1st Edition by Neeraj and Amarjit
2.	Construction Quality Management: Principles and Practice, Routledge.
3.	Estimating and Costing in Civil Engineering by B.N.Dutta
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc25_ce95/preview



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4102	PE	3	--	--	3	30	70	3 Hrs.
PRESTRESSED CONCRETE								
(For CE)								
Course Objectives:								
1.	To provide a fundamental understanding of Prestressed Concrete, including its basic concepts materials, advantages, and commonly used prestressing systems and methods.							
2.	To develop a comprehensive understanding of the analysis of Prestressed Concrete, including stress evaluation, load balancing concepts, and estimation of various prestress losses as per IS 1343:2012 provisions.							
3.	To develop the ability to analyze and design Prestressed Concrete members for flexure, shear, torsion, and axial forces by understanding failure mechanisms and code-based design procedures.							
Course Outcomes: After completing this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Explainthe principles, materials, and different prestressing systems and methods used in Prestressed Concrete and apply this knowledge in basic structural engineering contexts.							K2
2.	Analyze stresses in prestressed members and evaluate different types of prestress losses in accordance with IS 1343:2012 for practical structural applications.							K3
3.	Design Prestressed Concrete sections for flexure, shear, torsion, axial tension, compression, and bending using standard code provisions.							K3
4.	Analyze end blocks and anchorage zones of Prestressed Concrete members and design appropriate reinforcement based on stress distribution and code provisions.							K3
5.	Analyze continuous prestressed concrete beams and evaluate deflections, ensuring serviceability requirements are satisfied.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Prestressed Concrete: Introduction, Basic concepts of prestressing, need for high-strength steel and concrete, advantages of prestressed concrete. Materials for prestressed concrete, high-strength concrete and high-strength steel. Prestressing systems, Fressinet System, Gifford Udall, Magnel Blaton System. Pre-tensioning and post-tensioning systems.							
UNIT-II (10 Hrs)	Analysis of Prestress: Basic Assumptions, Pressure Line or Thrust Line, Concept of Load Balancing, cable profile, kern distance, Resultant Stresses at a Section, Stresses in Tendons as per IS 1343-2012. Prestressing losses, Loss Due to Elastic Deformation of Concrete, Loss due to shrinkage of Concrete, Loss Due to Creep of Concrete, Loss Due to Relaxation of stress in Steel, Loss							

	Due to Friction, Loss Due to Anchorage Slip, Loss Due to Curvature etc. I.S. code provisions.
UNIT-III (10 Hrs)	Flexural, shear, Torsional resistance and design of Prestressed concrete section: Types of flexural failure, code procedures, shear and principal stresses, prestressed concrete members in torsion, Design of sections for flexure, Axial Tension, Compression and bending, shear, Bond.
UNIT-IV (10 Hrs)	Analysis of end blocks: Transmission length, bond stress, transverse tensile stress, end zone reinforcement, flexural bond stress, IS code provisions for Bond and Transmission Length. Anchorage Zone in Post-Tensioned members introduction, Stress Distribution in End Block, Guyon's method of approach of analysis of end block (not more than 2 cables)
UNIT-V (10 Hrs)	Analysis of continuous beams: Elastic theory, Linear transformation and Concordant tendons. Deflections of pre-stressed concrete beams, Importance of control of deflections, factors influencing deflections, short term deflections of un-cracked member, prediction of long-term deflections.
Textbooks:	
1.	Prestressed Concrete, by N. Krishna Raju, McGraw Hill Pu
2.	Prestressed Concrete by G.S.Pandit and S.P. Gupta, CBS Publishers.
Reference Books:	
1.	Prestressed Concrete by S. Ramamrutham, Dhanpatrai Publishing Company References
2.	Prestressed Concrete Analysis and Design, Antone E.Naamane, TechnoPress3000
3.	Design of Prestressed Concrete-T.Y.Lin, Ned H. Burns, Wiley Publications
4.	Prestressed Concrete by N. Rajagopalan, Narosa Publishing
e-Resources	
1.	https://nptel.ac.in/courses/105106117
2.	https://nptel.ac.in/courses/105106686

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B23CE4103	PE	3	--	--	3	30	70	3 Hrs.
ADVANCED ENVIRONMENTAL ENGINEERING								
(For CE)								
Course Objectives:								
1.	To explain the importance of self-purification of streams and to derive Streeter- Phelps equation and to discuss the effects of various pollutants on receiving streams.							
2.	To introduce new concepts in biological treatment like nitrogen and phosphorous removal, anaerobic filters, rotating biological contactors, and U-tube aeration systems, their working principles and suitability.							
3.	To understand the characteristics and the treatment and disposal methods of liquid wastes produced in Dairy industry, Sugar industry and Pulp & paper industry.							
4.	To introduce various functional elements of urban solid waste management and to introduce various methods of solid waste treatment methods with special emphasis on recovery and reuse of solid waste.							
Course Outcomes At the end of the course, the student will be able to								
S. No	Outcome							Knowledge Level
1.	Determine dissolved oxygen balance in streams using Streeter–Phelps model.							K3
2.	Explain advanced biological wastewater treatment processes including nitrification, denitrification, and attached growth systems.							K2
3.	Summarize suitable treatment and disposal methods based on industrial wastewater characteristics.							K2
4.	Use principles of solid waste management and air pollutant classification for environmental management.							K3
5.	Explain air and noise pollution control methods based on their sources and environmental effects.							K2
SYLLABUS								
UNIT-I (10Hrs)	Self-purification in streams: Factors affecting self-purification; Dissolved Oxygen Balance in streams; Streeter-Phelps’s Dissolved Oxygen Model; Zones of Self-purification; Impact of pollutants on stream waters and usage of stream water with special reference to flora and fauna.							
UNIT-II (10 Hrs)	New Concepts in Biological Waste Treatment: Nitrogen removal by biological nitrification and de-nitrification; Phosphate removal from the activated sludge process; Rotating Biological Contactor; Anaerobic filters; U-Tube aeration systems.							
UNIT-III (10 Hrs)	Industrial Wastewater Treatment: Sugar Industry, Quantity of liquid waste; Characteristics of liquid waste; Methods of its treatment and disposal. Dairy Industry,							

	Quantity of liquid waste; Characteristics of liquid waste; Methods of its treatment and disposal. Pulp and Paper Industry, Quantity of liquid waste; Characteristics of liquid waste; Methods of its treatment and disposal.
UNIT-IV (10 Hrs)	Urban Solid Waste Management: Sources; Quantities and characteristics; Classification; Collection and transportation; Recovery and reuse; Treatment methods such as composting, incineration, sanitary landfill and pyrolysis. Sources and Classification of Air Pollution Stationary and mobile sources; Primary and secondary pollutants; Natural contaminants; Particulate matter; Aerosols; Gaseous pollutants. Meteorology and Air Pollution Atmospheric stability and temperature inversions; Maximum Mixing Depth; Wind direction and speed; Plume behaviour; Gaussian Dispersion Model; Plume rise; Wind rose.
UNIT-V (10 Hrs)	Effects of Air Pollution Global Effects: Global warming; Ozone depletion; Acid rains; Effects of air pollutants on human health; Effects on plants; Economical effects. Control of Air Pollution Objectives; Types of collection equipment: Settling chamber; Inertial separators; Cyclones; Filters; Electrostatic Precipitators; Scrubbers. Noise Pollution Introduction; Levels of noise; Noise rating systems; Measurement of noise; Sources of noise and their noise levels; Acceptable noise levels.
Textbooks:	
1.	Wastewater Treatment by M.N. Rao and A.K. Datta; Oxford & IBH Publishing Co. Pvt. Ltd., 2008.
2.	Environmental Pollution Control Engineering by C.S. Rao, New Age International, 2006.
Reference Books:	
1.	Wastewater Engineering: Treatment and Reuse – Metcalf & Eddy, McGraw-Hill
2.	Environmental Engineering – H.S. Peavy, D.R. Row & G. Tchobanoglous, McGraw-Hill
3.	Industrial Water Pollution Control – W.W. Eckenfelder Jr., McGraw-Hill
4.	Water and Wastewater Technology – Mark J. Hammer & Mark J. Hammer Jr., Pearson
5.	Integrated Solid Waste Management – George Tchobanoglous et al., McGraw-Hill
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc24_ge19/preview
2.	https://onlinecourses.nptel.ac.in/noc22_ch45/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4104	PE	3	--	--	3	30	70	3 Hrs.
URBAN HYDROLOGY								
(For CE)								
Course Objectives:								
1.	To Understand the importance of short duration rainfall runoff data for urban hydrology studies.							
2.	To Understand the concepts in design of various components of urban drainage systems.							
3.	To Understand the concepts of preparation master urban drainage system							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Apply the concept of the urban hydrologic cycle to evaluate the effects of urbanisation.							K3
2.	Develop intensity duration frequency curves for urban drainage systems							K3
3.	Develop design storms to size the various components of drainage systems							K3
4.	Apply best management practices to manage urban flooding							K3
5.	Prepare master drainage plan for an urbanized area							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Urbanisation and its effect on water cycle – urban hydrologic cycle – trends in urbanisation – Effect of urbanisation on hydrology							
UNIT-II (10 Hrs)	Precipitation Analysis: Importance of short duration of rainfall and runoff data, methods of estimation of time of concentration for design of urban drainage systems, Intensity, Duration, Frequency (IDF)curves, design storms for urban drainage systems							
UNIT-III (10 Hrs)	Approaches to urban drainage: Time of concentration, peak flow estimation approaches, rational method, NRCS curve number approach, runoff quantity and quality, wastewater and storm water reuse, major and minor systems. Elements of drainage systems: Open channel, underground drains, appurtenances, pumping, source control							
UNIT-IV (10 Hrs)	Analysis and Management: Stormwater drainage structures, design of stormwater network- Best Management Practices–detention and retention facilities, swales, constructed wetlands, models available for stormwater management							
UNIT-V (10 Hrs)	Master drainage plans: Issues to be concentrated upon – typical urban drainage master plan, interrelation between water resources investigation and urban planning processes, planning objectives, comprehensive planning, use of models in planning							

Textbooks:	
1.	Urban Hydrology by Hall M J (1984), Elsevier Applied Science Publisher
2.	Urban Hydrology – Quantity and Quality Analysis by Wanielista M P and Eaglin (1997), Wiley and Sons.
Reference Books:	
1.	Manual on Drainage in Urbanised areaby Geiger W. F., J Marsalek, W. J. Rawls and F. C. Zuidema, (1987 – 2 volumes), UNESCO.
2.	Hydrology – Quantity and Quality Analysis by Wanielista M P and Eaglin (1997), Wiley and Sons.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc24_ce10/preview
2.	https://archive.nptel.ac.in/courses/105/101/105101002/



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B23CE4105	PE	3	--	--	3	30	70	3 Hrs.
FOUNDATIONS ON EXPANSIVE SOILS								
(For CE)								
Course Objectives:								
1.	To understand the mineralogy, structure, and soil–water interaction of expansive soils.							
2.	To study the swelling behavior, including swell potential, swelling pressure, and heave characteristics of expansive soils.							
3.	To learn various foundation practices and design considerations for structures on expansive soils.							
4.	To understand different soil stabilization techniques and moisture control methods for improving expansive soils.							
5.	To study the shear strength behavior and stress state variables of partly saturated expansive soils.							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the mineralogical composition, soil structure, and soil–water interaction mechanisms influencing the behavior of expansive soils.							K2
2.	Apply concepts of swelling characteristics to determine swell potential, swelling pressure, and heave in expansive soils.							K3
3.	Apply suitable foundation techniques such as under-reamed piles and other methods for safe construction in expansive clay soils.							K3
4.	Apply soil stabilization methods including lime treatment, moisture barriers, and other admixtures to improve expansive soil performance.							K3
5.	Apply shear strength principles and suction measurement techniques to analyze the behavior of partly saturated expansive soils.							K3
SYLLABUS								
UNIT-I (10Hrs)	Fundamentals of clay minerology and soil water interaction:Clay mineralogy- Nature of soils-clay mineral structures- cation exchange- soil water-soil structure soil water interaction.							
UNIT-II (10 Hrs)	Swelling behaviour and heave characteristics of expansive soils: swelling characteristics - factors affecting swelling-swell potential - swelling pressure- methods of determination – factors affecting swell potential and swelling pressure – Heave factors affecting heave methods of determination of heave.							
UNIT-III (10 Hrs)	Foundation techniques and design in expansive clays :Foundation Practices in Expansive Clays – Sand cushion – Belled Piers – CNS layer technique – Under –reamed pile foundations – Construction techniques – design specifications – Load -carrying							

	capacity in compression and uplift of single and multi– under reamed piles in clays and sands–granular pile Anchors.
UNIT-IV (10 Hrs)	Soil stabilization and moisture control method: Lime soil columns and Lime slurry pressure injection – Stabilization with admixtures –Pre ponding – Vertical and Horizontal Moisture Barriers.
UNIT-V (10 Hrs)	Shear strength and suction characteristics of expansive soils: Shear strength of expansive soils-Katti's concept of bilinear strength envelope-Stress state variables in partly saturated soils- Fred lund's strength parameters-determination of matrix suction by axis translation technique-field suction measurement
Textbooks:	
1.	Katti, R.K.Search for Solutions to Problems in Black Cotton Soils, Central Board of Irrigation and Power (CBIP), New Delhi, 1980.
2.	Nelson, J.D., Miller, D.J., &Zornberg, J.G., Expansive Soils: Problems and Practice in Foundation and Pavement Engineering (Revised Edition), John Wiley & Sons, Inc., Hoboken, New Jersey, 2015.
Reference Books:	
1.	Holtz, R.D., Kovacs, W.D., and Sheahan, T.C., An Introduction to Geotechnical Engineering, 2nd Edition, Pearson Education, New Delhi, 2011.
2.	Das, B.M., and Sobhan, K., Principles of Geotechnical Engineering,8th Edition, Cengage Learning, Boston, 2014.
e-Resources	
1.	NPTEL Course: Soil Mechanics by Prof. B.V.S. Viswanadham, IIT Bombay https://nptel.ac.in/courses/105101084
2.	NPTEL Course: Foundation Engineering, IIT Roorkee by Prof. Mahendra Singh, Prof. N.K. Samadhiya, Prof. Priti Maheswari https://nptel.ac.in/courses/105107120

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4107	PE	3	--	--	3	30	70	3 Hrs.
GEOSYNTHETICS AND ITS APPLICATIONS								
(For CE)								
Course Objectives:								
1.	Introduce the different types of geosynthetics with emphasis on geotextiles and geogrids, their manufacturing methods, properties and classification.							
2.	Provide knowledge on geomembranes, geonets, geocomposites and geosynthetic clay liners along with their functions and basic characteristics.							
3.	Develop understanding of the application of geosynthetics in reinforced pavements and reinforced soil retaining structures.							
4.	Familiarize students with the use of geosynthetics in hydraulic and environmental engineering applications such as lining systems, landfills and erosion control.							
5.	Explain specifications, installation procedures, quality assurance, quality control and recent advancements in geosynthetics.							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the types, manufacturing methods, properties and classification of geotextiles and geogrids.							K2
2.	Describe geomembranes, geonets, geocomposites and geosynthetic clay liners along with their functions and basic properties.							K2
3.	Apply the concepts of geosynthetics in reinforced pavements and reinforced soil retaining walls.							K3
4.	Describe the use of geosynthetics in hydraulic and environmental applications such as lining systems, landfills and erosion control.							K2
5.	Explain specifications, installation procedures, quality assurance, quality control and emerging trends in geosynthetics.							K2
SYLLABUS								
UNIT-I (10Hrs)	Geotextiles and Geogrids: Definition and brief development of geosynthetics. Types and classification of geotextiles, raw materials and manufacturing methods. Types of geogrids, materials and manufacturing methods. Identification and classification of geotextiles and geogrids. Basic properties including physical, mechanical and hydraulic properties. Introduction to standards and testing methods (ASTM, ISO, BIS).							
UNIT-II (10 Hrs)	Geomembranes, Geonets and Geocomposites: Types of geomembranes, materials and manufacturing methods. Structure and applications of geonets. Types and functions of geocomposites. Introduction to geosynthetic clay liners. Basic properties including physical, mechanical, hydraulic and durability properties. Functions of geosynthetics such							

	as separation, filtration, drainage, reinforcement and containment.
UNIT-III (10 Hrs)	Reinforced Pavements and Retaining Structures: Application of geosynthetics in pavements, base reinforcement and basic concept of load distribution. Reinforced soil retaining walls: concept and components of reinforced soil retaining walls, construction methods (wrap-around and panel facings), design concepts
UNIT-IV (10 Hrs)	Geosynthetics in Hydraulic and Environmental Applications: Use of geomembranes in lining systems such as reservoirs, canals and ponds. Introduction to geosynthetic clay liners and their composition. Basic concepts of landfill systems including liners, covers and leachate collection. Erosion control methods using mats and geocells. Coastal and riverbank protection using geobags.
UNIT-V (10 Hrs)	Specification, Installation and Quality Aspects: Types of specifications including performance and prescriptive specifications (basic idea). Site preparation and installation procedures such as handling, seaming and anchoring. Seaming methods for geomembranes and basic seam testing methods. Introduction to Construction Quality Assurance (CQA) and Quality Control (CQC). Basic concepts of damage identification and repair, post-construction monitoring, and emerging trends such as recycled, biodegradable and smart geosynthetics.
Textbooks:	
1.	R.M. Koerner, Designing with Geosynthetics (6th Ed.), Xlibris Corporation, 2012
2.	B.M. Das. Principles of Geotechnical Engineering, Cengage Learning, 2019
Reference Books:	
1.	R.J. Bathurst (Ed.), Geosynthetics in Civil Engineering, Woodhead Publishing, 2007
e-Resources	
1.	NPTEL: Geosynthetics and Reinforced Soil Structures. https://nptel.ac.in/courses/105106052

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4108	PE	3	--	--	3	30	70	3 Hrs.
ENVIRONMENTAL IMPACT ASSESSMENT								
(For CE)								
Course Objectives:								
1.	To impart knowledge on different concepts of Environmental Impact Assessment.							
2.	To know procedures of risk assessment.							
3.	To learn the EIA methodologies and the criterion for selection of EIA methods.							
4.	To know pre-requisites for ISO 14001certification.							
5.	To know the procedures for environmental clearances and audit.							
6.	To appreciate the importance of stakeholder participation in EIA.							
Course Outcomes At the end of the course, the student will be able to								
S. No	Outcome							Knowledge Level
1.	Integrate EIA fundamentals, baseline data, and tools to develop assessment frameworks.							K3
2.	Distinguish and analyze suitability of EIA methodologies for impact assessment.							K3
3.	Develop impact prediction models and implement mitigation strategies.							K3
4.	Assess ecological impacts and analyze environmental risks and uncertainties.							K3
5.	Implement EIA procedures and audit mechanisms for environmental compliance.							K3
SYLLABUS								
UNIT-I (10Hrs)	Basic concepts of EIA: Elements of EIA, factors affecting EIA, Initial Environmental Examination, life cycle analysis preparation of Environmental Base map, Classification of environmental parameters, role of stakeholders in the EIA preparation, stages in EIA, Environmental economics, Cost/benefit Analysis, EIS and EMP. Identification of activities application of remote sensing and GIS for EIA.							
UNIT-II (10 Hrs)	EIA Methodologies: Introduction, Criteria for the selection of EIA Methodology, EIA methods, Ad-hoc methods, matrix methods, Network method Environmental Media Quality Index method, overlay methods. Impact of Developmental Activities and Land use, Introduction and Methodology for the assessment of soil and ground water, Delineation of study area.							
UNIT-III (10 Hrs)	EIA with reference to surface water, Air and Biological environment:Procurement of relevant soil quality, Impact prediction, Assessment of Impact significance, Identification and Incorporation of mitigation measures, Methodology for the assessment of Impacts on surface water environment, generalized approach for assessment of Air pollution Impact.							

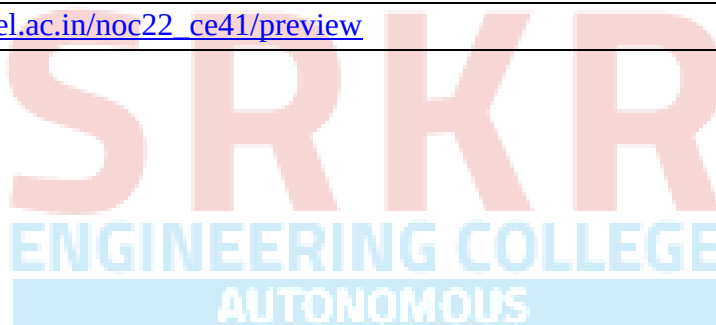
UNIT-IV (10 Hrs)	Environmental Risk Assessment and Risk management in EIA: Assessment of Impact of development Activities on Vegetation and wildlife, Environmental Impact of Deforestation. Risk assessment and treatment of uncertainty-key stages in performing an Environmental Risk Assessment- Advantages of Environmental Risk Assessment
UNIT-V (10 Hrs)	EIA Acts, Notifications and Guidelines: Provisions in the EIA notification, procedure for environmental clearance, and procedure for conducting environmental impact assessment report- evaluation of EIA report. Environmental legislation objectives, evaluation of Audit data and preparation of Audit report. Post Audit activities, Concept of ISO and ISO14000. Environmental compliance reports. Case studies and preparation of EIA statement for various Industries
Textbooks:	
1.	Environmental Impact Assessment, Canter Larry W., McGraw-Hill education Edi (1996)
2.	Environmental Impact Assessment Methodologies, Y. Anjaneyulu, B. S. Publication, Sultan Bazar, Hyderabad.
Reference Books:	
1.	Environmental Science and Engineering, J. Glynn and Gary W. Hein Ke Prentice Hall Publishers
2.	Environmental Science and Engineering, Suresh K. Dhaneja, S. K. Katania& Sons Publication, New Delhi.
3.	Environmental Pollution and Control, H. S. Bhatia, Galgotia Publication (P) Ltd, Delhi
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc26_ar13/preview?
2.	https://alison.com/course/principles-of-environmental-impact-assessment?

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4109	PE	3	--	--	3	30	70	3 Hrs.
DISASTER MANAGEMENT AND PREPAREDNESS								
(For CE)								
Course Objectives:								
1.	To provide an understanding of disasters, their types, causes, and the impact on communities, infrastructure, and the environment.							
2.	To familiarize with disaster management frameworks, policies, and practices adopted at local, national, and international levels.							
3.	To emphasize the role of civil engineering, planning, and modern technologies in disaster mitigation, preparedness, and resilient infrastructure development.							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the fundamental concepts of disaster, hazard, risk, vulnerability, and disaster management, and identify the disaster-prone areas in India.							K3
2.	Classify various types of natural and manmade disasters relevant to civil engineering, and apply basic engineering knowledge to recognize structural vulnerabilities.							K3
3.	Develop appropriate disaster mitigation plans and construct risk reduction strategies based on the analysis of disaster impacts on infrastructure, environment, and society.							K3
4.	Plan and organize community-based disaster preparedness strategies by applying the disaster management cycle in the context of civil engineering practices.							K3
5.	Utilize modern technologies like GIS, remote sensing, and IS codes to model disaster-resilient structures and infrastructure systems.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Disasters and Disaster Management: Definition and concepts: Disaster, Hazard, Risk, Vulnerability, Resilience, Classification of disasters: Natural and Manmade, Disaster management: Objectives and importance Relationship between disasters and human development, Disaster profile of India – vulnerability mapping, high-risk zones							
UNIT-II (10 Hrs)	Types of Disasters: Natural Disasters: Earthquakes: causes, seismic zoning in India, earthquake engineering basics, Floods and Flash Floods: river behaviour, urban flooding, floodplain zoning, Cyclones: wind loads, coastal vulnerability, Droughts: water scarcity, soil degradation, Landslides: causes, slope instability, mitigation structures Manmade Disasters: Structural failures, industrial accidents Fires, chemical and nuclear hazards, Transportation and infrastructure-related disasters.							

UNIT-III (10 Hrs)	Disaster Impacts and Risk Reduction Strategies: Impact on life, infrastructure, agriculture, environment, Livelihood loss, water source contamination, public health concerns, Disaster risk reduction (DRR): concepts, global and national frameworks (Sendai Framework, NDMA Guidelines), Structural and non-structural mitigation strategies in civil engineering, Vulnerability and capacity assessment.
UNIT-IV (10 Hrs)	Disaster Management Cycle and Community-Based Preparedness: Disaster management cycle: Prevention, Mitigation, Preparedness, Response, Recovery. Emergency planning and post-disaster reconstruction, Role of community in disaster risk reduction, Essentials of school and institutional safety programs, Community-based disaster management (CBDM) and resilience building, Role of social capital in disaster recovery.
UNIT-V (10 Hrs)	Technology and Tools in Disaster Management: Role of GIS and Remote Sensing in disaster assessment and planning, Early warning systems and disaster communication, Structural design codes and standards for disaster resilience (IS codes), Use of simulation, modelling, and multimedia in disaster education and response, Transformable indigenous knowledge and traditional practices in disaster risk reduction, Disaster management planning for critical infrastructure: roads, bridges, buildings, dams.
Textbooks:	
1.	Disaster Management – Global Challenges and Local Solutions’ by Rajibshah & RR Krishnamurthy (2009), Universities press.
2.	‘Disaster Management – Future Challenges and Opportunities’ by Jagbir Singh (2007), IK International Publishing House Pvt. Ltd.
Reference Books:	
1.	‘Disaster Management’ edited by HK Gupta (2003) Universities press.
2.	‘Disaster Science & Management’ by Tushar Bhattacharya, Tata McGraw Hill Education Pvt. Ltd., New Delhi.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/cec19_hs20/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4110	PE	3	--	--	3	30	70	3 Hrs.
TRAFFIC ENGINEERING AND MANAGEMENT								
(For CE)								
Course Objectives:								
1.	To provide an understanding of basic principles of traffic studies.							
2.	To familiarize how to focus on traffic, its properties, measurement, simulation and control. It starts with traffic flow variables and their measurement.							
3.	To emphasize the role of civil engineering, planning, and modern technologies in disaster mitigation, preparedness, and resilient infrastructure development.							
Course Outcomes At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Explain basic characteristics of traffic stream							K2
2.	Paraphrase traffic studies and analyze traffic data							K2
3.	Compute the capacity of highways							K3
4.	Solve traffic signal systems							K3
5.	Predict the transportation system & identify measures to improve situations.							K3
SYLLABUS								
UNIT-I (10Hrs)	Traffic Studies: Basic principles of Traffic, Volume, Speed and Density; Definitions and their interrelationships; Traffic Volume studies - Objectives, Methods of Volume counts, Presentation of Volume Data; Speed studies- Types of Speeds, Objectives, Methods of speed studies, Presentation of speed data. Delay Studies; Origin and Destination Studies.							
UNIT-II (10 Hrs)	Parking Studies: Parameters of parking, definitions, Parking inventory study, Parking survey by Patrolling method; Analysis of Parking Survey data; Accident studies- Causative factors of Road accidents, Accident data collection: Accident analysis and modeling; Road Safety Auditing, Measures to increase Road safety.							
UNIT-III (10 Hrs)	Capacity and LOS Analysis: Introduction to Traffic capacity, Analysis concepts, Level of Service, Basic definitions, Factors affecting Capacity and LOS, Capacity of Urban/Rural Highway, With or without access control, Computation of capacity and level of service for two lane highways.							
UNIT-IV (10 Hrs)	Intersections and Signal Designing:Type of Intersection, Conflicts at Intersection: At grade intersection, Uncontrolled, Channelization, signalized intersection, Types of Grades Separated Intersection. Fixed Time signals, Determination of Optimum Cycle length and Signal setting for Fixed Time signals, Warrants for Signals, Design of Isolated Traffic Signal by Webster method and IRC Method.							

UNIT-V (10 Hrs)	Transportation System Management: Measures for Improving vehicular flow, one-way Streets, Signal Improvement, Transit Stop Relocation, Parking Management, Reversible lanes- Reducing Peak Period Traffic - Strategies for working hours, Congestion Pricing, Differential Toll Policies.
Textbooks:	
1.	Traffic Engineering and Transportation Planning, L. R. Kadiyali, 4th Edition, Khanna Publishers, 1991.
2.	Highway Engineering, S. K. Khanna & C.E.G Justo, 9th Edition, Nem Chand & Bros Publisher, 2012.
Reference Books:	
1.	Principles of Highways Engineering and Traffic Analysis - Fred Mannering & Walter Kilareski, John Wiley & Sons Publication
2.	Traffic Engineering-Theory & Practice, Louis J. Pignataro, 2nd Edition, Prentice Hall Publication, 1973.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc22_ce41/preview



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4112	SEC	--	1	2	2	30	70	3 Hrs.
3D BUILDING MODELLING								
(For CE)								
Course Objectives:								
1	To introduce the concepts of Building Information Modelling (BIM) using Autodesk Revit.							
2	To familiarize students with Revit user interface and basic tools.							
3	To enable students to create architectural elements like walls, doors, windows, roofs, and stairs.							
4	To develop skills in modifying and editing building components.							
5	To provide hands-on experience in creating a complete building model.							
Course Outcomes: At the end of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1	ExplainBIM concepts and Revit interface							K2
2	Model walls and basic building components							K3
3	Model floors, roofs, ceilings, and curtain walls							K3
4	Design staircases, ramps, and railings							K4
5	Develop a complete building model using Revit tools							K5
SYLLABUS								
1	Introduction to BIM concepts and overview of Revit Architecture interface.							
2	Setting up project units, levels, and grids.							
3	Creation of walls using various draw tools and editing wall properties.							
4	Placement of doors, windows, and ventilators.							
5	Creation of curtain walls, stacked walls and architectural columns.							
6	Creation and modification of ceilings and roofs.							
7	Creation of staircases and types of stairs, ramps and railings.							
8	Creation of wall sweep and wall reveal, new materials for paint andcolour scheme.							
9	Finishing the model with colour fill legend, camera view, Hide elements and Render view.							
10	Massing and Site, Topo surface and Walk Through							
Reference Books:								
1	Autodesk Revit 2024 Architecture, Elise Moss, SDC Publications, 2023							
2	Mastering Autodesk Revit 2024, Robert Yori and Marcus Kim, Sybex (Wiley), 2023.							
3	Design Integration Using Autodesk Revit 2024, Daniel John Stine, SDC Publications, 2023.							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	--	--
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives:								
1.	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.							
2.	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.							
3.	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.							K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.							K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.							K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grassroots governance.							K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.							K3
SYLLABUS								
UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution, Sources and constitutional history, Features, Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (6 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							
UNIT-III	State Government and its Administration Governor: Role and Position, CM and							

(6 Hrs)	Council of ministers, State Secretariat: Organisation, Structure and Functions.
UNIT-IV (6 Hrs)	Local Administration: District's Administration Head, Role and Importance, Municipalities, Mayor and role of Elected Representative, CEO of Municipal Corporation Panchayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy, (Different departments), Village level, Role of Elected and Appointed officials, Importance of grass root democracy.
UNIT-V (6 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissioner State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd. New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

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SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation:R23		IV/ IV-B.Tech. II-Semester							
CIVILENGINEERING									
COURSESTRUCTURE (Witheffectfrom2023-24admittedBatchonwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CE4201	Internship and Project	PR	--	--	24	12	60	140	200
TOTAL			0	0	24	12	60	140	200



SRKR

ENGINEERING COLLEGE

AUTONOMOUS

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE4201	PR	--	--	24	12	60	140	3 Hrs.
PROJECT WORK								
(For CE)								
Course Objectives:								
1.	To provide an opportunity to work in group on a topic / problem / experimentation							
2.	To encourage creative thinking process							
3.	To provide an opportunity to analyze and discuss the results to draw conclusions							
4.	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify a current problem through literature/field/case studies							K3
2.	Identify the objectives and methodology for solving the problem							K3
3.	Design and Develop technology/process for solving the problem							K4
4.	Evaluate the technology/process							K5
*The object of Project Work is to enable the student to take up investigative study in the broad field of Civil Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work.								
The assignment to normally include:								
a) Survey and study of published literature on the assigned topic.								
b) Working out a preliminary approach to the problem relating to the assigned topic.								
c) Conducting preliminary Analysis/Modeling/Simulation/Experiment/Design/ Feasibility.								
d) Preparing a written report on the study conducted for presentation to the department.								
e) Final Seminar, as oral Presentation before a departmental committee.								